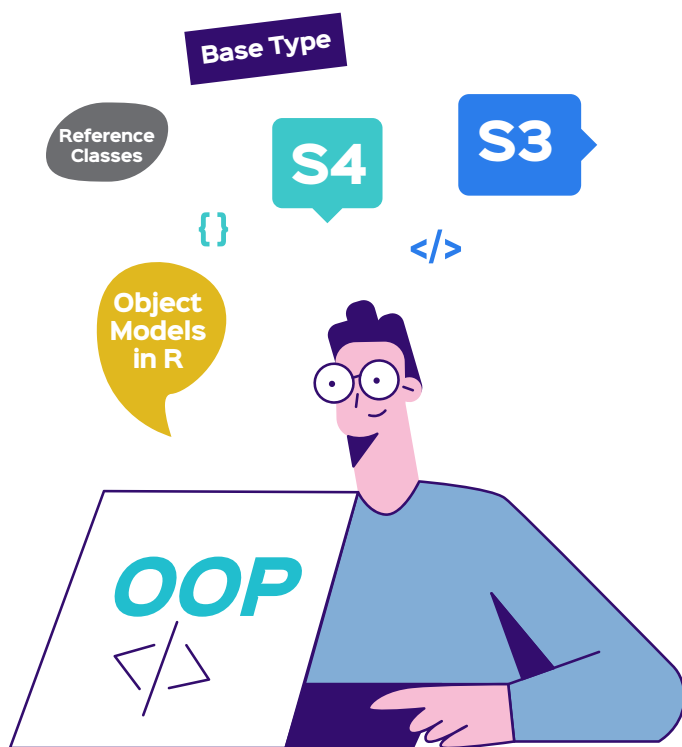


Object Oriented Programming in R

R is a functional programming language, but it does have support for object-oriented programming (OOP). In this thirteenth article in the R series, we shall learn about S3 and S4 classes (recommended) in R.



The basic classes for built-in types are listed below.

Object	Class
builtin	function
character	character
closure	function
complex	complex
double	numeric
environment	environment
expression	expression
integer	integer
language	call
list	list
logical	logical
NULL	NULL
pairlist	pairlist
special	function
symbol	name

Many software packages like 'lattice' and 'ggplot2' have been written using R objects. We will use the R version 4.1.2 installed on Parabola GNU/Linux-libre (x86-64) for the code snippets.

```
$ R --version
R version 4.1.2 (2021-11-01) -- "Bird Hippie"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)
```

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S3 class

The S3 object is an R primitive object with attributes and a class name. Consider the *mtcars* data set in the lattice library:

```
> library(lattice)
> head(mtcars)

      mpg  cyl  disp  hp drat   wt  qsec vs am gear carb
Mazda RX4   21.0   6  160 110 3.90 2.620 16.46  0  1   4    4
Mazda RX4 Wag 21.0   6  160 110 3.90 2.875 17.02  0  1   4    4
Datsun 710   22.8   4  108  93 3.85 2.320 18.61  1  1   4    1
Hornet 4 Drive 21.4   6  258 110 3.08 3.215 19.44  1  0   3    1
Hornet Sportabout 18.7  8 360 175 3.15 3.440 17.02  0  0   3    2
Valiant    18.1   6  225 105 2.76 3.460 20.22  1  0   3    1
```